

SHEARWATER, NS, Canada

WMO: 716010

Lat: 44.6297N

Lon: 63.5133W

Elev: 44

StdP: 100.80

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.7	-13.3	-21.0	0.6	-14.4	-18.7	0.7	-12.0	13.9	0.0	12.6	-0.2	4.9	310	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	26.5	19.7	24.9	18.8	23.5	18.2	21.3	24.4	20.4	23.1	19.6	22.0	4.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	15.0	22.6	19.4	14.2	21.7	18.6	13.5	20.8	62.2	24.7	58.9	23.3	56.0	22.1	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
11.4	10.0	8.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-18.7	30.5	3.1	1.8	-21.0	31.7	-22.8	32.8	-24.5	33.8	-26.7	35.0		
			-19.3	23.3	2.9	0.9	-21.4	24.0	-23.1	24.5	-24.8	25.0	-26.9	25.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-3.7	-3.3	-0.1	4.7	9.6	14.4	18.6	19.0	16.0	10.5	5.2	0.0
	DBStd	8.76	5.35	4.74	4.12	3.15	2.98	3.17	2.33	2.20	3.10	3.56	4.17	4.71
	HDD10.0	1816	423	372	314	161	44	3	0	0	1	38	152	309
	HDD18.3	3996	682	605	572	409	272	124	25	17	83	244	395	567
	CDD10.0	952	0	0	1	2	31	134	266	280	179	52	7	0
	CDD18.3	89	0	0	0	0	1	5	32	39	12	1	0	0
	CDH23.3	362	0	0	1	0	9	44	144	126	37	1	0	0
	CDH26.7	53	0	0	0	0	1	8	23	14	7	0	0	0
(14) Wind	WSAvg	4.1	4.9	4.8	4.8	4.5	3.9	3.5	3.0	3.1	3.5	4.1	4.5	4.7
(15) Precipitation	PrecAvg	1414	132	114	123	110	111	103	101	96	97	128	139	151
	PrecMax	1795	271	229	243	208	214	349	229	320	225	256	258	253
	PrecMin	983	42	34	45	19	26	32	13	25	27	32	38	67
	PrecStd	198	47	47	46	42	45	55	48	61	46	52	55	51
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	9.1	13.9	17.2	24.3	27.8	29.5	28.5	26.6	21.7	15.9	12.3
		MCWB	10.1	7.8	9.1	10.0	15.6	18.9	21.2	20.4	20.3	16.3	14.2	11.5
	2%	DB	8.3	6.8	9.0	13.8	19.4	24.1	26.4	26.1	23.9	18.9	14.4	10.5
		MCWB	7.7	5.7	6.1	8.4	12.8	17.3	19.8	19.6	18.8	16.1	13.1	9.8
	5%	DB	6.2	4.9	7.1	11.5	16.7	22.0	24.8	24.7	22.0	17.3	13.1	8.9
		MCWB	5.4	4.0	5.1	7.5	11.4	16.4	18.9	18.9	18.1	15.1	12.1	8.2
	10%	DB	4.0	3.3	5.5	9.7	14.6	20.0	23.2	23.2	20.6	16.0	11.7	7.1
		MCWB	3.1	2.3	3.8	6.8	10.2	15.4	18.3	18.4	17.5	14.1	10.6	6.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	8.2	9.9	12.0	16.9	21.0	22.3	22.2	22.0	18.6	15.1	11.9
		MCDB	10.7	8.7	11.5	14.4	21.2	25.5	26.8	25.7	24.6	20.0	15.6	12.3
	2%	WB	7.9	6.0	7.1	10.0	14.2	18.9	21.2	21.2	20.5	17.1	13.6	10.0
		MCDB	8.2	6.6	8.0	11.8	17.6	22.2	24.9	23.8	22.4	18.3	14.1	10.4
	5%	WB	5.6	4.2	5.6	8.6	12.5	17.5	20.2	20.3	19.3	15.8	12.3	8.3
		MCDB	6.1	4.7	6.7	10.3	15.1	20.2	23.1	22.9	20.9	16.8	12.9	8.8
	10%	WB	3.2	2.4	3.9	7.3	11.1	16.3	19.3	19.5	18.4	14.4	10.7	6.3
		MCDB	3.9	3.1	5.2	9.3	13.5	18.8	21.6	22.0	19.9	15.7	11.5	6.9
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	7.5	7.1	7.3	7.9	8.1	7.7	7.7	7.7	7.0	6.4	6.7
		MCDBR	9.8	8.4	9.1	10.7	11.8	11.8	10.5	9.8	9.0	8.1	7.4	8.6
	5% WB	MCWBR	10.1	8.2	7.4	7.2	7.2	6.8	5.4	5.2	5.6	6.1	7.3	9.0
		MCWBR	10.0	8.4	8.5	8.8	9.9	10.1	8.6	7.8	7.0	7.1	7.1	8.3
(40) Clear-Sky Solar Irradiance	taub	taub	0.302	0.296	0.323	0.348	0.369	0.398	0.414	0.395	0.353	0.332	0.315	0.296
		taud	2.473	2.521	2.476	2.449	2.437	2.381	2.358	2.409	2.500	2.555	2.552	2.537
	Ebn at Noon	Ebn at Noon	822	903	916	915	901	872	852	858	872	848	803	798
		Edh at Noon	70	79	95	106	111	118	119	110	92	77	65	60
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.43	2.33	3.53	4.31	4.95	5.38	5.35	5.00	4.07	2.64	1.52	1.13
	RadStd	RadStd	0.11	0.17	0.24	0.35	0.64	0.50	0.39	0.35	0.27	0.27	0.16	0.09

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	+0.37	+0.50	N/A	N/A	+56	+17

Nomenclature: See separate page